

Day 4: Hands-on Subnetting Practical

Network given: **192.168.10.0/24** | Task: divide into 4 equal subnets

Task 1: Divide 192.168.10.0/24 into 4 Equal Subnets

New subnet mask: **/26 = 255.255.255.192**. Each subnet has 64 addresses (62 usable hosts).

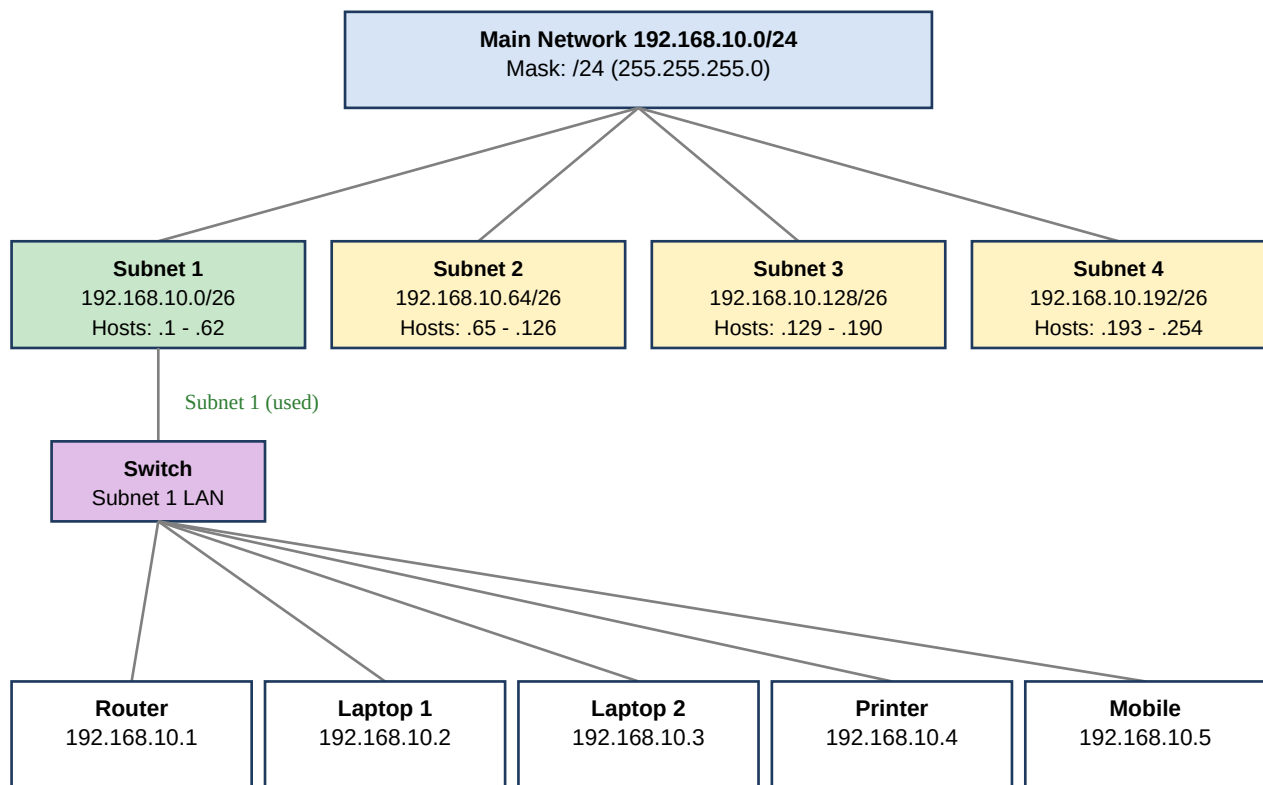
Subnet	Network Address	Subnet Mask	Usable Host Range	Broadcast Address
Subnet 1	192.168.10.0/26	255.255.255.192	192.168.10.1 - 192.168.10.62	192.168.10.63
Subnet 2	192.168.10.64/26	255.255.255.192	192.168.10.65 - 192.168.10.126	192.168.10.127
Subnet 3	192.168.10.128/26	255.255.255.192	192.168.10.129 - 192.168.10.190	192.168.10.191
Subnet 4	192.168.10.192/26	255.255.255.192	192.168.10.193 - 192.168.10.254	192.168.10.255

Task 2: IP Assignment in Subnet 1 (192.168.10.0/26)

Device	IP Address	Subnet Mask	Note
Router	192.168.10.1	255.255.255.192	Default gateway (first usable IP)
Laptop 1	192.168.10.2	255.255.255.192	Host
Laptop 2	192.168.10.3	255.255.255.192	Host
Printer	192.168.10.4	255.255.255.192	Host
Mobile	192.168.10.5	255.255.255.192	Host

Important: 192.168.10.0 is the network address and 192.168.10.63 is the broadcast address, so neither is given to any device. Gateway for all devices: 192.168.10.1.

Task 3: Subnet Plan Diagram



All 5 devices connect to the switch and share the same subnet (192.168.10.0/26).

Green = Subnet 1 (in use). Yellow = Subnets 2, 3, 4 (free for future use, e.g. other departments).

Task 4: Step-by-Step Explanation (for a beginner friend)

Step 1: Understand what /24 means

192.168.10.0/24 means the first 24 bits are the **network part** and the last 8 bits are the **host part**.
Mask: 255.255.255.0. With 8 host bits we have $2^8 = 256$ addresses (0 to 255).

Step 2: How many bits to borrow for 4 subnets?

We need 4 subnets. Formula: $2^n = \text{number of subnets}$. $2^2 = 4$, so **n = 2 bits** must be borrowed from the host part.

Step 3: Find the new subnet mask

New prefix = $24 + 2 = \mathbf{/26}$. In binary the last octet of the mask becomes $11000000 = 128 + 64 = \mathbf{192}$.
So the mask is **255.255.255.192**.

Step 4: Hosts per subnet

Host bits left = $32 - 26 = 6$. Total addresses = $2^6 = 64$. Usable hosts = $64 - 2 = \mathbf{62}$ (we subtract the network and broadcast addresses).

Step 5: Find the block size (jump)

Block size = $256 - 192 = \mathbf{64}$. So subnets start at 0, 64, 128, 192 (each jump adds 64).

Step 6: Network, broadcast and host range

Network address = start of each block. Broadcast = (next network address - 1). First host = network + 1, last host = broadcast - 1. Example for Subnet 1: network 192.168.10.0, broadcast 192.168.10.63, hosts .1 to .62. Next: 64 to 127, 128 to 191, 192 to 255.

Step 7: Assign IPs to devices

We choose Subnet 1. Router gets the first usable IP (192.168.10.1) as the gateway. Then Laptop 1 = .2, Laptop 2 = .3, Printer = .4, Mobile = .5. All are between .1 and .62, so they are valid. We never use .0 or .63.

Step 8: Verify

Check that all 5 IPs are in the same subnet: each lies in 192.168.10.0 to 192.168.10.63 and uses mask 255.255.255.192. Devices in the same subnet can talk directly; to reach other subnets they go through the router (192.168.10.1).

Quick formulas: Subnets = 2^n | Hosts per subnet = $2^h - 2$ | Block size = 256 - mask octet value